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zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ id devuser
uid=1001(devuser) gid=1001(devuser) groups=1001(devuser)
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ grep devuser /etc/passwd
devuser:x:1001:1001:~/home/devuser:/bin/bash
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ ls /home/devuser
ls: cannot open directory '/home/devuser': Permission denied
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ sudo ls -la /home/devuser
[sudo] password for zainwright:
total 20
drwxr-x--- 2 devuser devuser 4096 Sep 11 17:31 .
drwxr-xr-x 4 root    root    4096 Sep 11 17:31 ..
-rw-r--r-- 1 devuser devuser 220 Mar 31 2024 .bash_logout
-rw-r--r-- 1 devuser devuser 3771 Mar 31 2024 .bashrc
-rw-r--r-- 1 devuser devuser 807 Mar 31 2024 .profile
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ pwd
/home/zainwright/cvnp1601-portfolio/week03-provisioning
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ grep devuser /etc/passwd
devuser:x:1001:1001:~/home/devuser:/bin/bash
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ pwd
/home/zainwright/cvnp1601-portfolio/week03-provisioning
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$

```

It is important to use -aG instead of just -G. -aG will still carry over the previous memberships along with the newly specified one. It is likely that many memberships get replaced by the specified one without notice. Thankfully if you do notice, you can restore it with usermod -aG.

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zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ sudo groupadd developers
[sudo] password for zainwright:
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ id devuser
uid=1001(devuser) gid=1001(devuser) groups=1001(devuser)
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ sudo usermod -aG developers devuser
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ id dev user
id: 'dev': no such user
id: 'user': no such user
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ id devuser
uid=1001(devuser) gid=1001(devuser) groups=1001(devuser),1002(developers)
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ sudo usermod -G developers devuser
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ id devuser
uid=1001(devuser) gid=1001(devuser) groups=1001(devuser),1002(developers)
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ |

```

/etc/shadow is separate from /etc/passwd for security. Shadow contains a lot of sensitive password info that should be well hidden. Passwd is more for basic account information that is sensitive.

```

zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ grep devuser /etc/passwd
devuser:x:1001:1001:~/home/devuser:/bin/bash
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ sudo grep devuser /etc/shadow
[sudo] password for zainwright:
devuser:$y$j9T$XAfHcEkLHRHCB7jXbN8oV1$LrosXlpjqG6h7vX54GUwVxuZw1JqjiRPXAi3by/ighB:20707:0:99999:7:::
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ grep developers /etc/group
developers:x:1002:devuser
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$

```

Visudo was needed to do a quick rule edit. We had to have a rule for devuser to be able to restart the service without root privileges.

```
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ sudo visudo
zainwright@cvnp1601-lab:~/cvnp1601-portfolio/week03-provisioning$ sudo -u devuser bash
devuser@cvnp1601-lab:/home/zainwright/cvnp1601-portfolio/week03-provisioning$ sudo -l
[sudo] password for devuser:
Sorry, try again.
[sudo] password for devuser:
Sorry, try again.
[sudo] password for devuser:
Matching Defaults entries for devuser on cvnp1601-lab:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin, use_pty

User devuser may run the following commands on cvnp1601-lab:
    (root) /usr/bin/systemctl nginx
devuser@cvnp1601-lab:/home/zainwright/cvnp1601-portfolio/week03-provisioning$ sudo /usr/bin/systemctl s
top nginx
sudo: /usr/bin/systemctl: command not found
devuser@cvnp1601-lab:/home/zainwright/cvnp1601-portfolio/week03-provisioning$ sudo /usr/bin/systemctl
stop nginx
Sorry, user devuser is not allowed to execute '/usr/bin/systemctl stop nginx' as root on cvnp1601-lab.
devuser@cvnp1601-lab:/home/zainwright/cvnp1601-portfolio/week03-provisioning$ |

devuser@cvnp1601-lab:/home/zainwright/cvnp1601-portfolio/week03-provisioning$ sudo -u devuser sudo -l
Sorry, user devuser is not allowed to execute '/usr/bin/sudo -l' as devuser on cvnp1601-lab.
devuser@cvnp1601-lab:/home/zainwright/cvnp1601-portfolio/week03-provisioning$ sudo -u devuser sudo -l
Sorry, user devuser is not allowed to execute '/usr/bin/sudo -l' as devuser on cvnp1601-lab.
devuser@cvnp1601-lab:/home/zainwright/cvnp1601-portfolio/week03-provisioning$ |
```